

Project Development Phase Model Performance Test

Date	16 February 2026
Team ID	LTVIP2026TMIDS54574
Project Name	Smart Sorting Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Metrics	Regression Model: Not Applicable (Project uses Classification Model) Classification Model: Confusion Matrix – Shows correct and incorrect predictions of Fresh and Rotten classes Accuracy Score – 96% Classification Report – Precision: 0.96, Recall: 0.95, F1-score: 0.95	Accuracy : 0.9038354641467482 Precision: 0.8917127071823204 Recall : 0.9149659863945578 F1 Score : 0.9031897034135422 Confusion Matrix
2.	Tune the Model	Hyperparameter Tuning: • Learning Rate = 0.001 • Batch Size = 32 • Number of Epochs = 10–15 • Optimizer = Adam • Loss Function = CrossEntropyLoss • Transfer Learning Model = ResNet18 / MobileNet Validation Method: • Train-Test Split (80% Training, 20% Testing) • Validation performed on test dataset to evaluate accuracy	Training Accuracy vs Epochs

